

1. Offline mining

Once per model and skill library

Known-task activation pairs



Remove shared rank-one component U_0



Low-rank skill bases U_t and anchors μ_t

U_0

library

Language-model weights remain frozen

2. New-task adaptation

Once per task; at most four examples

New task demonstrations



Contrastive extraction
de-shared signature z



Weighted group LASSO
support S and coefficients c_t

z

S, c_t

Signals: $\|z\|$ and reconstruction residual

Reconstruction is a diagnostic of library coverage

3. Query serving

Reuse the cached task operator

Query hidden state h



Demonstration direction
+ gated geometric correction
gate g uses z and the anchor



Selected bases and weighted anchors
projector P_S , reference μ_S

Greedy decoding from the intervened state

No demonstration tokens in the steering query



Separately evaluated serving policies

Weak $\|z\|$ → prompt-state replacement | Large residual → full ICL with demonstrations